

学术论文撰写

Abstract (摘要)

寸土寸金，拔高质量

A.常规写法

- 一句话描述研究背景
- 一两句话描述现有工作，并突出本文先前工作差异
- 三四句话描述技术流程，让别人知道本文方法的整体流程、关键模块等信息(高度凝练)
- 一两句话表述结果，鼓励给出代码GitHub地址

B.开门见山写法

- 一句话直接描述本文首次做了什么
- 简要解释动机来源

Abstract (摘要)

- **Abstract—** Anatomical and functional image fusion is an important technique in a variety of medical and biological applications. Recently, deep learning (DL)-based methods have become a mainstream direction in the field of multi-modal image fusion. However, existing DL-based fusion approaches often struggle to effectively capture both local features and global contextual information simultaneously. In addition, the scale diversity of features — a crucial issue in image fusion — has not received sufficient attention in most existing works. In this paper, to address the above problems, we propose a MixFormer-based multi-scale network, termed MM-Net, for anatomical and functional image fusion. In our method, an improved MixFormer backbone is introduced to sufficiently extract both local features and global contextual information at multiple scales from the source images. The features from different source images are fused at multiple scales using a multi-source spatial attention-based cross-modality feature fusion (CMFF) module. The scale diversity of the fused features is further enriched by a series of multi-scale feature interaction (MSFI) and feature aggregation upsampling (FAU) modules. Moreover, a loss function consisting of both spatial-domain and frequency-domain components is devised to train the proposed fusion model. Experimental results demonstrate that our method outperforms several state-of-the-art fusion methods in both qualitative and quantitative evaluations, and the proposed model exhibits strong generalization capability. The source code of our fusion method will be made available at <https://github.com/yulin316316>.
- **Abstract—** In this letter, a sparse representation (SR) model named convolutional sparsity-based morphological component analysis (CS-MCA) is introduced for pixel-level medical image fusion. Unlike the standard SR model, which is based on a single image component and overlapping patches, the proposed CS-MCA model can simultaneously achieve multi-component and global SRs of source images by integrating morphological component analysis (MCA) and convolutional sparse representation (CSR) into a unified optimization framework. For each source image, in the proposed fusion method, the CSRs of its cartoon and texture components are first obtained by the CS-MCA model using pre-learned dictionaries. Then, for each image component, the sparse coefficients of all the source images are merged, and the fused component is reconstructed using the corresponding dictionary. Finally, the fused image is obtained as the superposition of the fused cartoon and texture components. Experimental results demonstrate that the proposed method outperforms several benchmark and state-of-the-art SR-based fusion methods in terms of both visual perception and objective evaluation metrics.

Introduction (引言)

决定文章70%命运

是一个讲“故事”的地方

- 背景意义

不要整句copy他人论文，避免查重不过

- 研究现状

高度凝练，脉络清晰，不要列举细节(放Related work中)

Introduction (引言)

- 存在问题(最关键的部分)

阐述当前研究存在的问题，问题和本文贡献要清晰对应，问题要让人信服，考虑使用图、表等方式直观体现问题

- 本文工作

针对上述问题，阐述本文核心思想，体现出创新性，描述方法整体框架和主要元素

- 总结贡献(重要)

合理总结，按重要性排序(总-分等方式)，反复打磨

Related Work (相关工作)

本文工作与已有工作的联系和区别

- 对研究方向相关工作进行分类阐述
 - 力求清晰，避免混乱
 - 力求全面，避免遗漏
 - 力求客观，避免主观
- 在每一类方法描述结束后，要针对性总结
 - 体现现有工作的不足，突出本文工作的区别与特色

Method (方法)

清晰、严谨、公式化地展现提出的方法

- 如果引入的技术非常新，可以加入**预热**部分
- 描述网络结构一般**先总体框架**，再逐一阐述子模块
- 描述每个模块前，说清**动机**由来(引用他人工作)
- 描述每个模块后，说清**好处**(引后文实验结果)
- 每一幅**图都非常重要**，尤其是**总体框图**

Method (方法)

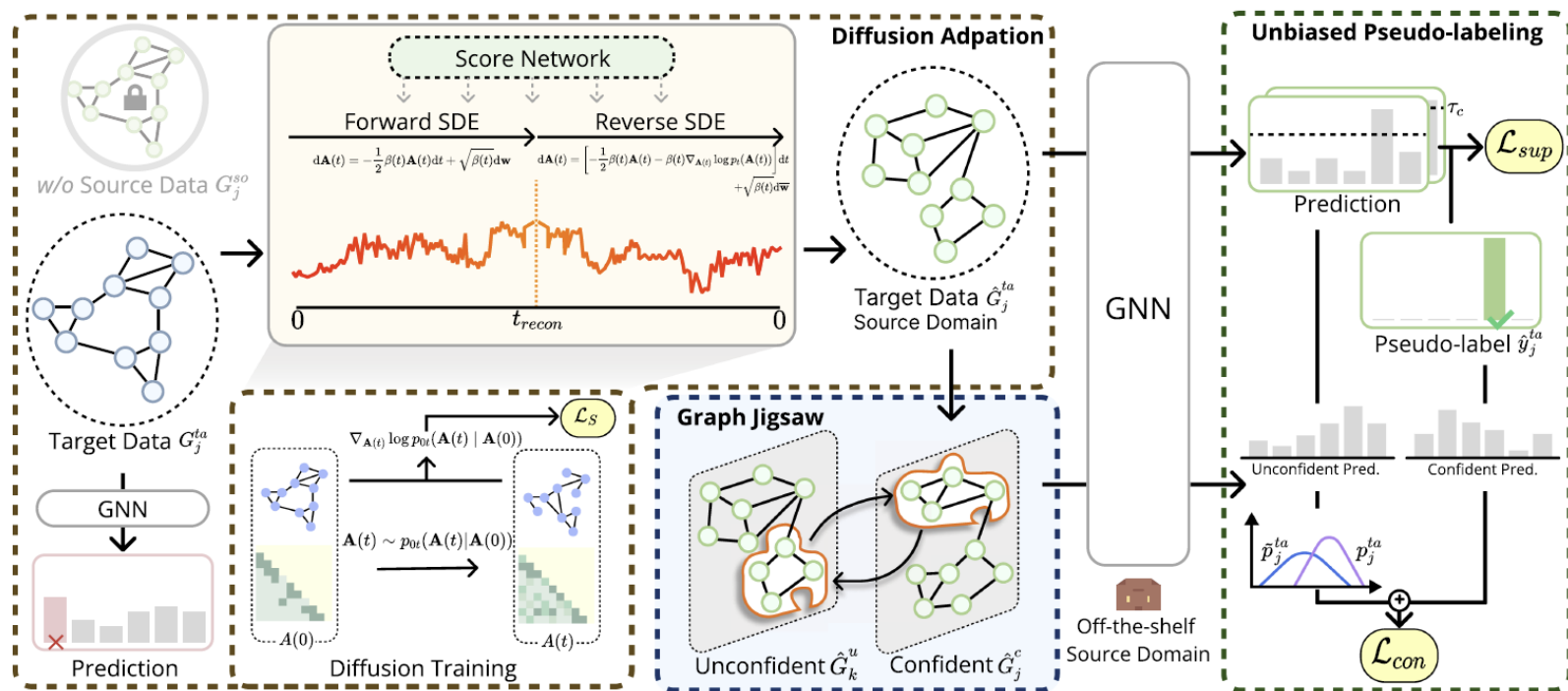


Fig. 2. Overview of GALA. We employ source graphs to train a score-based graph diffusion model and then transform target graphs into source-style graphs. Moreover, we introduce adaptive class-specific thresholds to generate confident graphs with pseudo-labels and then utilize graph jigsaw to exchange subgraphs between graph pairs for consistency learning.

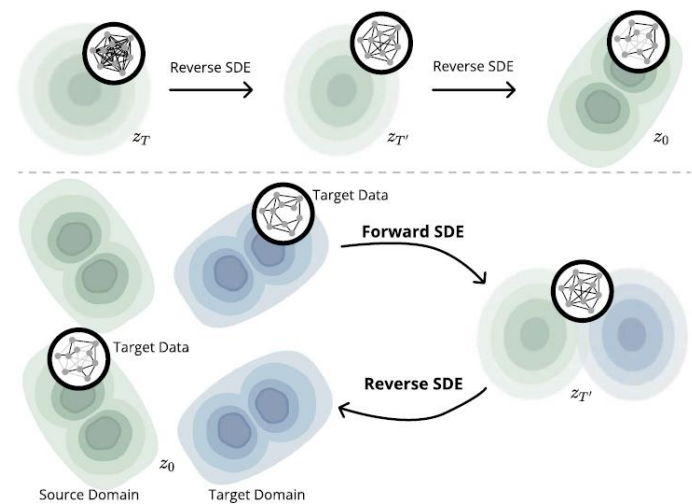


Fig. 1. A brief motivation of GALA. Previous score-based diffusion models usually generate new data by sampling from a prior distribution (upper), while ours GALA transforms the target graphs back to the source domain (lower).

Discussion (讨论)

依据实验结果对方法进行更深层次的讨论

- 尽量加入**定量**的数据，不要只是定性
- 适当运用**图、表**展示，不要只是文字
- 适当讨论方法的**局限性**(limitation)
- 合理引出**未来工作**

Conclusion (结论)

总结全文、首尾呼应

- 再一次总结贡献以及方法达到的效果
- 不要直接copy摘要或引言里的语句(调整表达)
- 如果没有Discussion, conclusion内容要适当加强

Reference (参考文献)

格式、格式、格式!

- 注重格式，姓名、期、卷、页、大小写(态度问题)
- 适当引用**所投期刊会议**的文献(特别是近两年的)
- 多引用**近三年**的最新工作(当前年的也要包括)
- 涵盖该方向**知名同行**的工作(很可能是审稿人)

评审意见分析

抓住审稿人的心理

- 不要着急开始修改论文，静心思考一两天(新的启发)
- 综合分析多个审稿人的意见(共性问题要更重视)
- 制订详细的修改和回复方案(包括时间进度)
- 善于从审稿意见中挖掘信息(审稿人身份等)

论文修改及回复信撰写

回复信更加重要

- 论文和回复信**迭代式**提升(两轮左右)
- 回复信要**全面详尽**(审稿人可能只关注回复信)
- 回复信要**层次清晰**(使用不同颜色、字体)
- 修改稿要顾及发表的合理性(相关内容可放在回复信)

论文修改及回复信撰写

RESPONSE TO REVIEWERS AND COVER LETTER

We here submit our revised article "*SelfGCN: Graph Convolution Network with Self-Attention for Skeleton-based Action Recognition*." We are very grateful for the comments that we have received from the reviewers on the original version TIP-30135-2023, especially since they were quite positive and helpful. We are also thankful for this chance to improve our paper. We have tried our best to address all comments. In the following text, we number all comments sequentially, provide our detailed answers, and highlight the corresponding improvements that we have applied to the paper.

The most important change is that we have added four sets of ablation experiments, targeting different structural designs in SelfGCN (serial, multiplication, parallel), the recognition of local and global actions in the graph convolution module and the Self-Attention module, the feature fusion of the graph convolution module and the Self-Attention module, and a more lightweight design of SelfGCN. Through these ablation experiments, we found that our innovative parallel design of graph convolution and Self-Attention in SelfGCN can achieve model decoupling. This allows the graph convolution module and the Self-Attention module to complement each other in capturing local and global features, respectively, without affecting each other's learning focus. Additionally, the results of ablation experiments confirm that the cross-branch interactions in MFSG do enhance feature learning and contribute positively to action recognition. However, the parameter cost of these interactions is also evident. Therefore, we have additionally conducted preliminary research on a more lightweight design of SelfGCN.

Furthermore, to better illustrate the differences and connections between SelfGCN and the related works, we visualized the spatial modeling in graph convolution in the revised manuscript. We systematically optimized the overall framework of SelfGCN, providing a more detailed presentation of the interrelation between its various modules and the stacked network.

In summary, we found and confirmed that our method performs better than the state of the art. It is also robust with respect to the number of channels in the source data. We improved our text with respect to the shortcomings pointed out by the reviewers. Additionally, to make our work easier to re-use and re-reproduce, we have made our project publicly available at <https://github.com/SunPengP/SelfGCN>.

3. REVIEWER #1

Comment 3. In this paper, the authors introduce the Graph Convolution Network with Self-Attention (SelfGCN) to enhance skeleton-based action recognition. They emphasize the effectiveness of GCNs but also point out their limitations in capturing long-range node relationships due to a primary focus on short-range dependencies. Furthermore, existing methods typically employ a uniform skeleton topology for all frames, restricting feature learning capabilities. To tackle these challenges, the authors propose SelfGCN, comprising two key modules: the *Mixing Features across Self-Attention and Graph Convolution* (MFSG) module and the *Temporal-Specific Spatial Self-Attention* (TSSA) module. The MFSG module simultaneously applies graph convolution and self-attention to model local and global relationships between joints, using a bidirectional interactive learning strategy that combines channel and spatial dimensions. The TSSA module utilizes self-attention to capture spatial relationships within each frame of a skeleton sequence, considering the unique spatial features of individual frames. The authors conducted extensive experiments on benchmark datasets, demonstrating the superior performance of their method compared to many recent state-of-the-art models in the past three years. However, certain concerns are raised regarding the novelty of this approach, which may severely limit the overall technical contribution, as elaborated below.

Answer 3. Thank you for this correct and positive summary of our work and the valuable comments on the problems in our work. We carefully implemented the comments you gave, as listed in the following points. We did indeed not emphasize the novelty of our method in the context of the existing works sufficiently well. Therefore, in the new paper version, we improve the discussion of the novel contributions of our work, see, for example, our answers to your [Comments 4 and 9](#) as well as [Comment 27](#) of Reviewer 4.

Comment 4. The authors claim to be the first to combine graph convolution with self-attention. However, this contribution claim is not entirely accurate, as existing methods like GAT (Graph Attention Network) have already combined graph convolution with self-attention. To ensure precision in their contribution claims, the authors should consider acknowledging these existing methods and possibly include them in a comparative analysis, possibly as part of their ablation study.

古今做学问者必经的三个境界

- 昨夜西风调碧树，独上高楼，望尽天涯路
- 衣带渐宽终不悔，为伊消得人憔悴
- 众里寻他千百度，暮然回首，那人却在灯火阑珊处

学问苦乐，日久自悟

与各位共勉